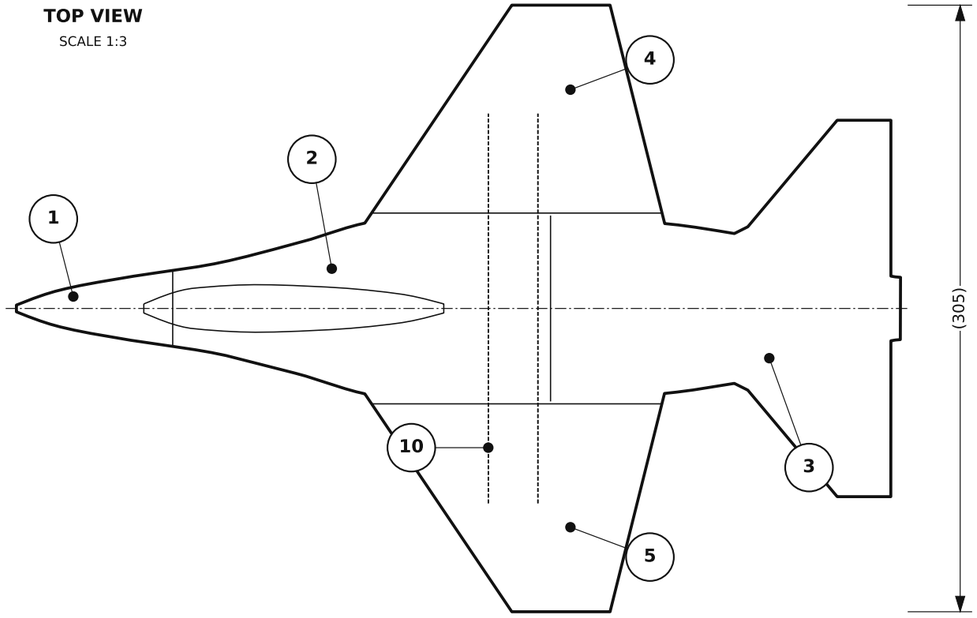
REV	DESCRIPTION	DATE
A	INITIAL RELEASE (PROTOTYPE)	2026-10-11

+Y = RH WING (ITEM 04)

TOP VIEW

SCALE 1:3





FRONT VIEW

SCALE 1:3

PARTS LIST

ITEM	PART NO.	DESCRIPTION	MATERIAL	QTY	MASS
1	F35-RC-01	NOSE CONE, CRUMPLE	PLA	1	13 g
2	F35-RC-02	FUSELAGE, FORWARD	PLA	1	72 g
3	F35-RC-03	FUSELAGE, AFT + TAILS	PLA	1	75 g
4	F35-RC-04	WING PANEL, RH	PLA	1	24 g
5	F35-RC-05	WING PANEL, LH (OPP. 04)	PLA	1	24 g
6	COTS	INSERT, HEAT-SET M3 x 5 (D 4.2)	BRASS	5	
7	COTS	INSERT, HEAT-SET M2 x 3 (D 3.0)	BRASS	4	
8	COTS	SCREW, M3 x 6 BUTTON HEAD	STEEL	5	
9	COTS	SCREW, M2 x 5 PAN HEAD	STEEL	2	
10	COTS	ROD, CARBON ⌀3.0 x 195	CFRP	2	
11	TBD	SERVO, 9 G CLASS + 2X M2 x 6	-	1	

DATUM REFERENCE FRAMES (EACH BUILT FROM THE MATING FEATURES)

PART	DATUM A	DATUM B	DATUM C	MATES
01	REAR FACE X80	SPIGOT OUTER SURF.	3X INSERT HOLES	02 SOCKET
02	AFT FACE X270	AFT SPIGOT SURF.	2X SPAR BORES	01, 03, 04/05
03	FWD FACE X270	SOCKET INNER SURF.	2X CLEAR. HOLES	02 SPIGOT
04/05	ROOT FACE	FWD SPAR BORE	AFT SPAR BORE	02 FACE D

GENERAL NOTES / TOLERANCE STACK:

1. MODEL-BASED: CAD MODEL IS MASTER; UNDIMENSIONED GEOMETRY IS BASIC.
2. JOINTS (01-02, 02-03): 0.25 RADIAL CLEARANCE NOMINAL. SPIGOT PROFILE 0.15 + SOCKET PROFILE 0.20 -> MIN CLEARANCE 0.25-0.075-0.10 = 0.075.
3. FIXED-FASTENER RULE H - F = T1 + T2: M3 ⌀3.40-3.00 = 0.40 = 0.15 (INSERT, PROJECTED 2) + 0.25 (CLEARANCE). M2: 0.30 = 0.15 + 0.15.
4. SPAR RODS FLOAT IN ⌀3.20 BORES: H - F = 0.20 >= EACH PART'S POSITION TOL.
5. INSPECT: CALIPERS, PIN GAUGES 2.8/3.2/4.0, TRIAL FIT OF EACH JOINT.

DRAWING INDEX:

SHEET 1 F35-RC-00 ASSEMBLY, PARTS LIST, DATUM SCHEME
SHEET 2 F35-RC-01 NOSE CONE, CRUMPLE (REPLACEABLE)
SHEET 3 F35-RC-02 FUSELAGE, FORWARD
SHEET 4 F35-RC-03 FUSELAGE, AFT + TAILS + SERVO BAY
SHEET 5 F35-RC-04/-05 WING PANELS RH / LH

UNLESS OTHERWISE SPECIFIED:

DIMENSIONS ARE IN MILLIMETERS

INTERPRET PER ASME Y14.5-2018

TOLERANCES (FDM PRINTED PART):

X ± 0.5

X.X ± 0.3

X.XX ± 0.10

ANGLES ± 1°

BREAK SHARP EDGES 0.2 MAX

SURFACES WITHOUT PROFILE CALLOUT:

PROFILE OF SURFACE 0.8 TO CAD MODEL

F-35 RC AIRFRAME | FRUCD / A. RODRIGUEZ

CAD: Onshape "F35 shape" > Part Studio 1 > feature "F-35 RC Airframe" (FeatureScript, default parameters)

AIRFRAME ASSEMBLY, DRAWING INDEX + DATUM SCHEME

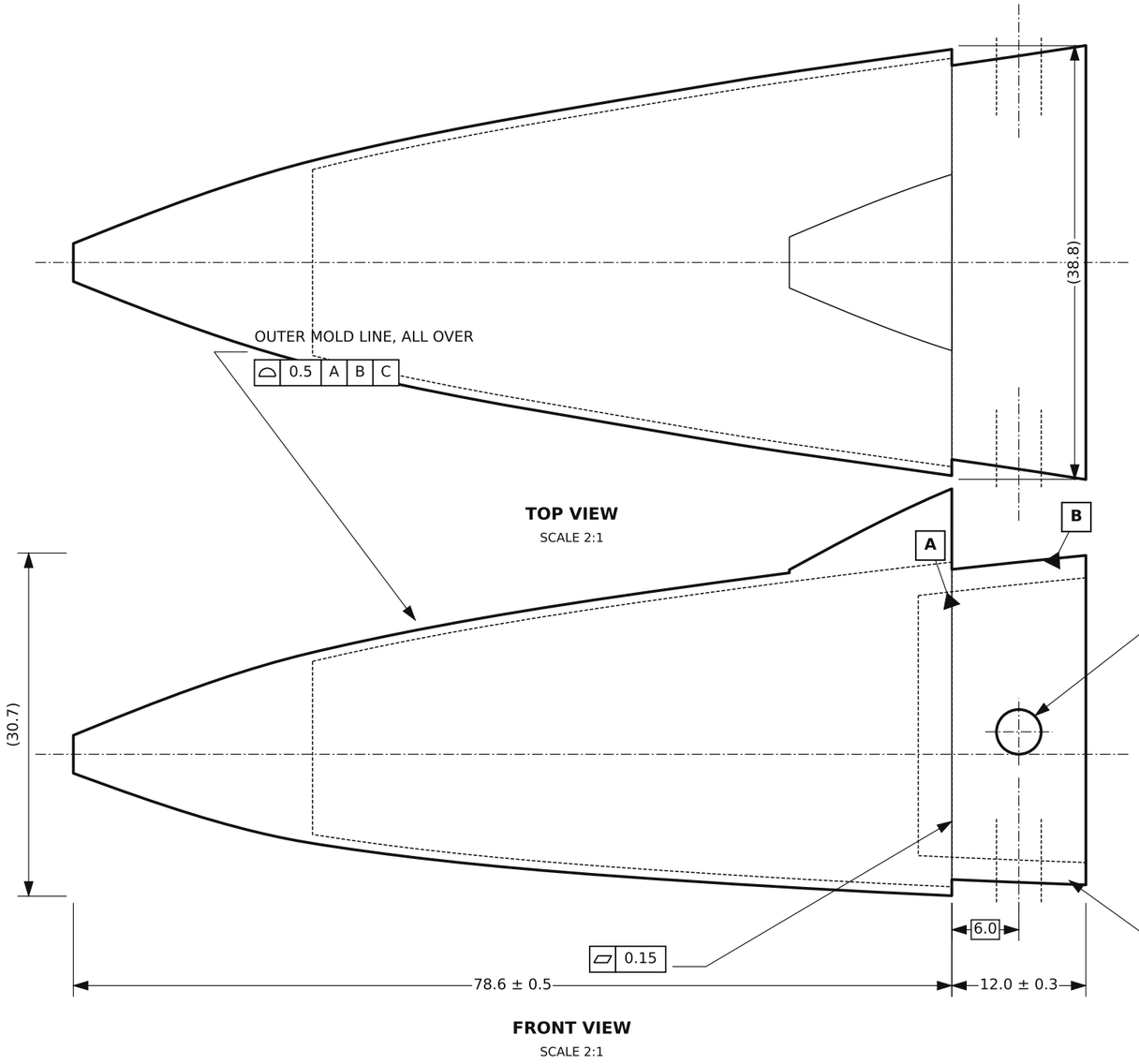
Top-level assembly: 5 printed parts, heat-set insert joints, carbon spar rods. Sheets 2-5 detail each part.

PART NO.	REV	SCALE	SHEET
F35-RC-00	**A**	**1:3**	**1 OF 5**

MATERIAL	EST. MASS	UNITS / STANDARD
PLA (FDM, 1.75 mm)	208 g (PRINTED)	mm | ASME Y14.5-2018

DRAWN	CHECKED / APPROVED	PROJECTION
A. RODRIGUEZ 2026-10-11	—	3RD ANGLE

REV	DESCRIPTION	DATE
A	INITIAL RELEASE (PROTOTYPE)	2026-10-11



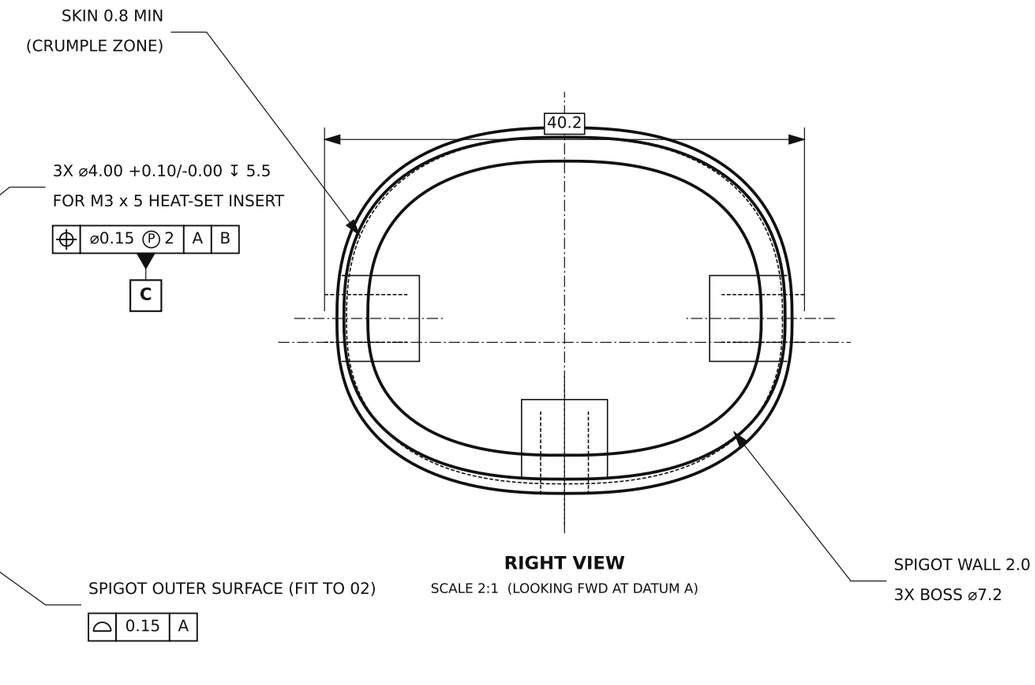
TOP VIEW
SCALE 2:1

FRONT VIEW
SCALE 2:1

NOTES:

1. PRINT NOSE-UP, STANDING ON DATUM A (SPIGOT DOWN). 0.12 MM LAYERS, 2 WALLS, 0% INFILL.
2. THE 0.8 MM SKIN IS THE CRUSH ELEMENT: NO INFILL, RIBS OR GLUE INSIDE.
3. INSTALL 3X M3 x 5 HEAT-SET INSERTS RADIALLY FROM OUTSIDE, 220-230 °C, FLUSH TO -0.2 BELOW SPIGOT SURFACE.
4. DATUM C = 3X INSERT PATTERN (CLOCKING ABOUT DATUM B).
5. SURFACES AND BASIC LOCATIONS NOT SHOWN ARE DEFINED BY THE CAD MODEL.
6. CONSUMABLE: REPLACE AFTER ANY IMPACT THAT CRACKS OR BUCKLES THE SKIN.

SKIN-TO-SKIN AT HOLE AXES (X = 86.0)



RIGHT VIEW
SCALE 2:1 (LOOKING FWD AT DATUM A)

UNLESS OTHERWISE SPECIFIED:

DIMENSIONS ARE IN MILLIMETERS

INTERPRET PER ASME Y14.5-2018

TOLERANCES (FDM PRINTED PART):

X ± 0.5

X.X ± 0.3

X.XX ± 0.10

ANGLES ± 1°

BREAK SHARP EDGES 0.2 MAX

SURFACES WITHOUT PROFILE CALLOUT:

PROFILE OF SURFACE 0.8 TO CAD MODEL

F-35 RC AIRFRAME | FRUCD / A. RODRIGUEZ

CAD: Onshape "F35 shape" > Part Studio 1 > feature "F-35 RC Airframe" (FeatureScript, default parameters)

NOSE CONE, CRUMPLE (REPLACEABLE)

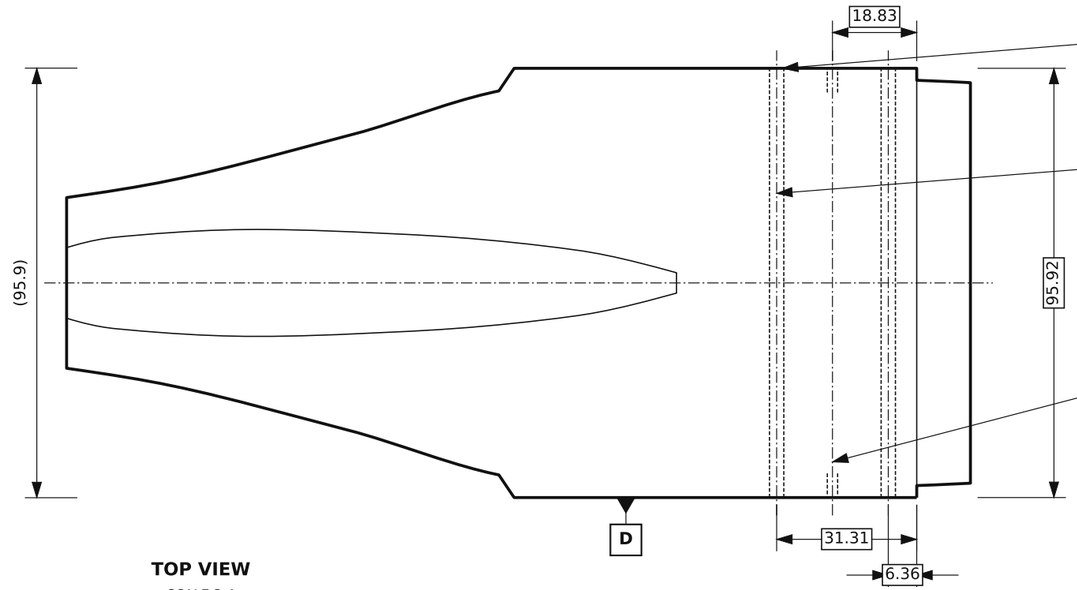
Sacrificial energy-absorbing nose; slides into item 02 and is retained by 3 radial M3 screws.

PART NO.	REV	SCALE	SHEET
F35-RC-01	**A**	**2:1**	**2 OF 5**

MATERIAL	EST. MASS	UNITS / STANDARD
PLA (FDM, 1.75 mm)	13 g	mm | ASME Y14.5-2018

DRAWN	CHECKED / APPROVED	PROJECTION
A. RODRIGUEZ 2026-10-11	—	3RD ANGLE

REV	DESCRIPTION	DATE
A	INITIAL RELEASE (PROTOTYPE)	2026-10-11



2X WING MOUNT FACE (Y = ±47.96)

▱	0.10	
⊥	0.20	A

2X ⌀3.20 +0.10/-0.00 THRU (SPAR BORES, CARBON ROD ⌀3 x 195)

⌖	⌀0.20	A	B

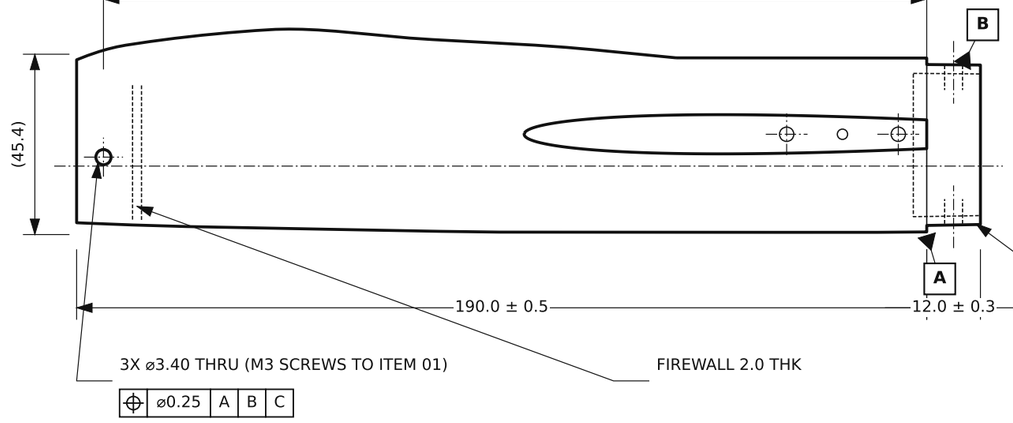
C

2X ⌀2.30 THRU STUB (M2 WING SCREW)

⌖	⌀0.15	A	B	C

TOP VIEW

SCALE 3:4



SPIGOT OUTER SURFACE (FIT TO ITEM 03)

⌓	0.15	A

3X ⌀3.40 THRU (M3 SCREWS TO ITEM 01)

⌖	⌀0.25	A	B	C

FIREWALL 2.0 THK

FRONT VIEW

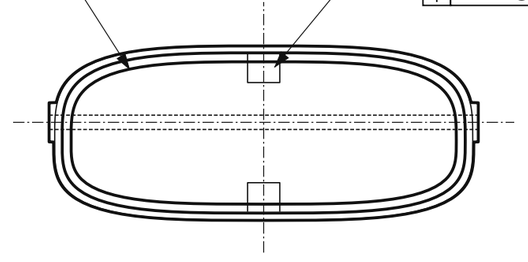
SCALE 3:4

OUTER MOLD LINE, ALL OVER

⌓	0.5	A	B	C

2X ⌀4.00 +0.10/-0.00 ↧ 5.5
M3 x 5 INSERT, TOP & BOTTOM

⌖	⌀0.15	Ⓟ 2	A	B	C



RIGHT VIEW

SCALE 3:4 (LOOKING FWD AT DATUM A)

NOTES:

1. PRINT ON DATUM A FACE (SPIGOT UP); TREE SUPPORT UNDER CANOPY ONLY. 0.16 MM LAYERS, 3 WALLS, 0% INFILL; 4 WALLS AROUND SPAR BORES.
2. DATUM C = 2X SPAR BORE PATTERN. DATUM D = RH WING MOUNT FACE.
3. NOSE SOCKET (X 80-92): INNER SURFACE PROFILE 0.20 |A|B|C|; FRONT FACE PARALLEL 0.30 TO A.
4. INSTALL 2X M3 x 5 INSERTS FROM OUTSIDE (TOP, BOTTOM OF SPIGOT).
5. SURFACES / BASIC LOCATIONS NOT SHOWN: PER CAD MODEL.

UNLESS OTHERWISE SPECIFIED:

DIMENSIONS ARE IN MILLIMETERS

INTERPRET PER ASME Y14.5-2018

TOLERANCES (FDM PRINTED PART):

X ± 0.5

X.X ± 0.3

X.XX ± 0.10

ANGLES ± 1°

BREAK SHARP EDGES 0.2 MAX

SURFACES WITHOUT PROFILE CALLOUT:

PROFILE OF SURFACE 0.8 TO CAD MODEL

F-35 RC AIRFRAME | FRUCD / A. RODRIGUEZ

CAD: Onshape "F35 shape" > Part Studio 1 > feature "F-35 RC Airframe" (FeatureScript, default parameters)

FUSELAGE, FORWARD (CANOPY + WING CARRY-THROUGH)

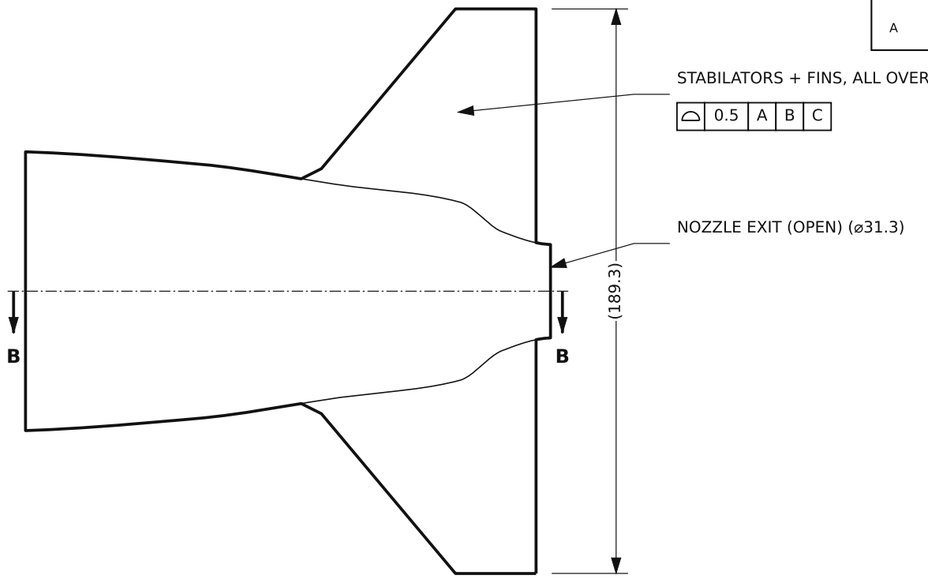
Carries canopy, firewall, both wing roots (2 spar bores) and the aft spigot joint to item 03.

PART NO.	REV	SCALE	SHEET
F35-RC-02	**A**	**3:4**	**3 OF 5**

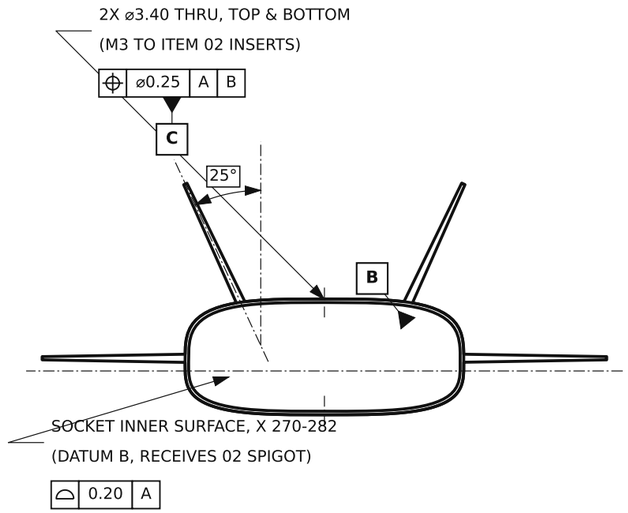
MATERIAL	EST. MASS	UNITS / STANDARD
PLA (FDM, 1.75 mm)	72 g	mm | ASME Y14.5-2018

DRAWN	CHECKED / APPROVED	PROJECTION
A. RODRIGUEZ 2026-10-11	—	3RD ANGLE

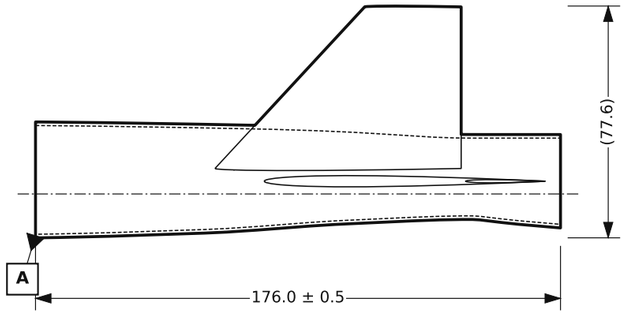
REV	DESCRIPTION	DATE
A	INITIAL RELEASE (PROTOTYPE)	2026-10-11



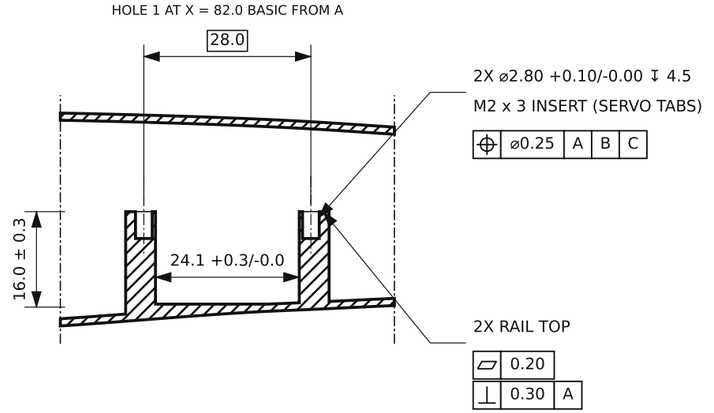
TOP VIEW
SCALE 1:2



LEFT VIEW
SCALE 1:2 (LOOKING AFT AT DATUM A)



FRONT VIEW
SCALE 1:2



SECTION B-B
SCALE 1:1 (SERVO BAY)

NOTES:

1. PRINT ON DATUM A FACE, NOZZLE UP; TREE SUPPORTS UNDER STABILATORS. 0.16 MM LAYERS, 3 WALLS, 0% INFILL; TAILS 2 WALLS + 15% GYROID.
2. DATUM C = 2X CLEARANCE HOLES (TOP/BOTTOM) AT 6.0 BASIC FROM A.
3. SERVO MOUNT SIZED FOR 9 G CLASS SERVO (23.5 x 12.5 BODY, 28 HOLE PITCH); RESIZE VIA FEATURE PARAMETERS BEFORE PRINTING A LARGER SERVO.
4. FIN CANT 25° BASIC, SYMMETRIC ABOUT VERTICAL CENTER PLANE.
5. SURFACES / BASIC LOCATIONS NOT SHOWN: PER CAD MODEL.

UNLESS OTHERWISE SPECIFIED:

DIMENSIONS ARE IN MILLIMETERS

INTERPRET PER ASME Y14.5-2018

TOLERANCES (FDM PRINTED PART):

X ± 0.5

X.X ± 0.3

X.XX ± 0.10

ANGLES ± 1°

BREAK SHARP EDGES 0.2 MAX

SURFACES WITHOUT PROFILE CALLOUT:

PROFILE OF SURFACE 0.8 TO CAD MODEL

F-35 RC AIRFRAME | FRUCD / A. RODRIGUEZ

CAD: Onshape "F35 shape" > Part Studio 1 > feature "F-35 RC Airframe" (FeatureScript, default parameters)

FUSELAGE, AFT (TAILS + SERVO BAY)

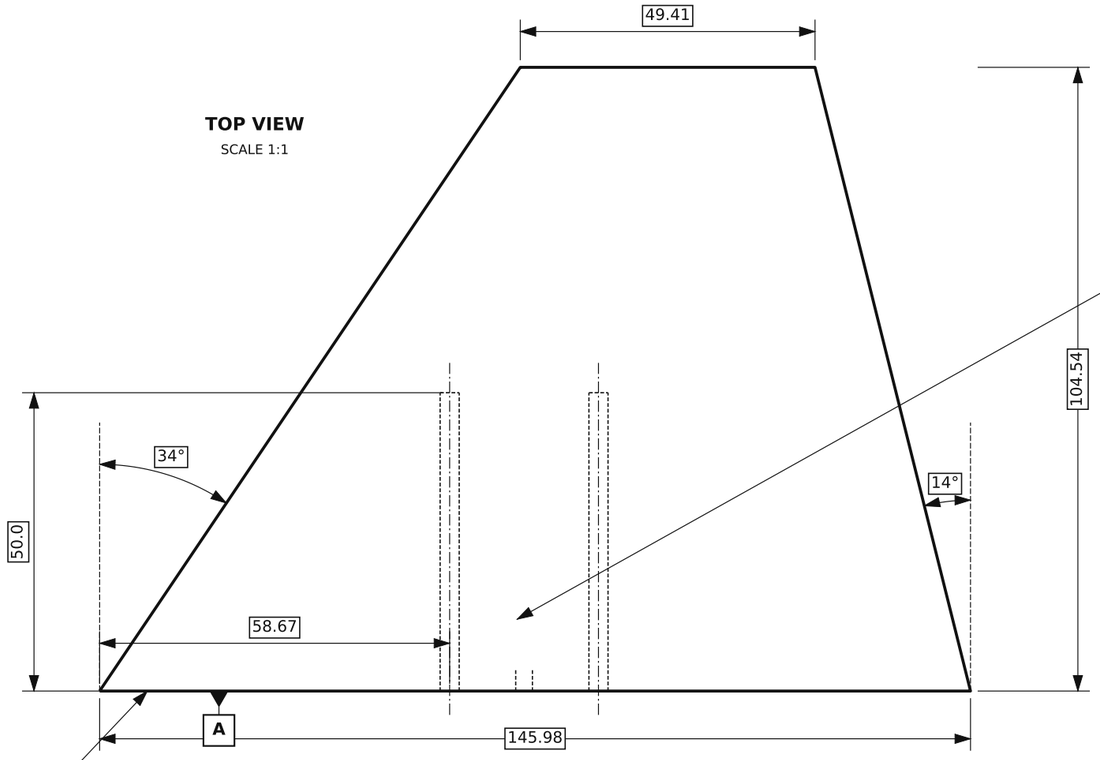
Receives item 02 spigot; carries stabilators, canted fins, open nozzle and the servo mount.

PART NO.	REV	SCALE	SHEET
F35-RC-03	**A**	**1:2**	**4 OF 5**

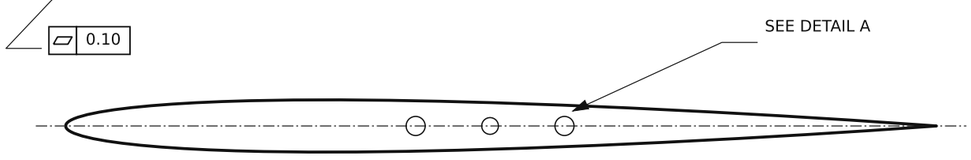
MATERIAL	EST. MASS	UNITS / STANDARD
PLA (FDM, 1.75 mm)	75 g	mm | ASME Y14.5-2018

DRAWN	CHECKED / APPROVED	PROJECTION
A. RODRIGUEZ 2026-10-11	—	3RD ANGLE

REV	DESCRIPTION	DATE
A	INITIAL RELEASE (PROTOTYPE)	2026-10-11



TOP VIEW
SCALE 1:1



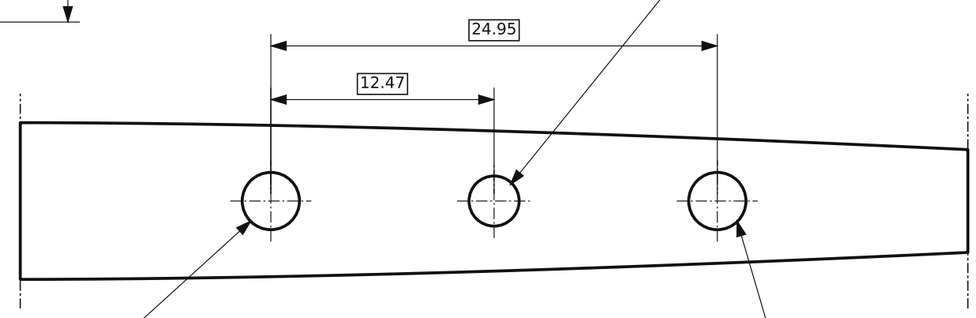
FRONT VIEW (ROOT FACE)
SCALE 1:1

NACA 0006 UPPER & LOWER SURFACE

⌓	0.30	A	B	C

NOTES:

1. ITEM 05 (LH) IS THE MIRROR IMAGE OF ITEM 04 ABOUT DATUM A'S MATING PLANE; ALL REQUIREMENTS APPLY.
2. PRINT STANDING ON DATUM A (ROOT FACE), 5 MM BRIM. 0.12 MM LAYERS, 2 WALLS, 8-10% GYROID, OUTER WALL FIRST, SEAM ON TRAILING EDGE.
3. 4 WALLS AROUND SPAR BORES. MIN WALL AT BORE BOTTOM 1.3.
4. TRAILING EDGE: 0.4 MIN AS PRINTED; DO NOT SAND THROUGH.
5. ASSEMBLY: SLIDE ON 2X CARBON RODS (ITEM 10) UNTIL DATUM A SEATS ON ITEM 02 FACE D; RETAIN WITH M2 x 5 SCREW FROM INSIDE FUSELAGE.
6. WING INCIDENCE 0° AND DIHEDRAL 0° ARE SET BY DATUMS B, C.



DETAIL A (ROOT FACE, DATUM A)
SCALE 3:1

ALL HOLES ON CHORD PLANE (BASIC)

UNLESS OTHERWISE SPECIFIED:

DIMENSIONS ARE IN MILLIMETERS

INTERPRET PER ASME Y14.5-2018

TOLERANCES (FDM PRINTED PART):

X ± 0.5

X.X ± 0.3

X.XX ± 0.10

ANGLES ± 1°

BREAK SHARP EDGES 0.2 MAX

SURFACES WITHOUT PROFILE CALLOUT:

PROFILE OF SURFACE 0.8 TO CAD MODEL

F-35 RC AIRFRAME | FRUCD / A. RODRIGUEZ

CAD: Onshape "F35 shape" > Part Studio 1 > feature "F-35 RC Airframe" (FeatureScript, default parameters)

WING PANEL, RH (05 = LH, OPPOSITE HAND)

NACA 0006 wing; locates on 2 carbon spar rods (datums B, C) against the fuselage mount face (datum A).

PART NO.	REV	SCALE	SHEET
F35-RC-04 / -05	**A**	**1:1**	**5 OF 5**

MATERIAL	EST. MASS	UNITS / STANDARD
PLA (FDM, 1.75 mm)	24 g EA	mm | ASME Y14.5-2018

DRAWN	CHECKED / APPROVED	PROJECTION
A. RODRIGUEZ 2026-10-11	—	3RD ANGLE